

3.2.3 How technology and cultural changes can impact on the work of designers

Socioeconomic influences

Content	Potential links to maths and science
Socioeconomic influences have helped to shape product design and manufacture, including: • trends • street culture music and media • world events, eg WW1, WW2 • the rise of youth culture and anti-authoritarian attitudes • the influence of workwear garments on fashion • the role of women in society • sport and leisure • technological developments • music, film, royalty and celebrities.	

Major developments in technology

Content	Potential links to maths and science
How major developments in technology are shaping product design and manufacture, including: • the introduction of regenerated and synthetic fibres during the 20th century • the development of fabric finishes, e-textiles and smart materials • new methods of manufacturing clothing and textile materials including mass production as opposed to bespoke, automated manufacturing including CAD and CAM • new decorative techniques such as laser printing • development in the care of textiles.	An awareness of scientific advancements/ discoveries and their potential development.

Product life cycle

Content	Potential links to maths and science
The stages of the product life cycle, including: • design introduction • evolution • growth • maturity • decline • replacement. Students should be able illustrate their understanding with examples of how, with reference to specific products, designers have refined and redeveloped products.	

Social, moral and ethical issues

Content	Potential links to maths and science
The responsibilities of designers and manufacturers, including: • products are made using sustainable materials and ethical production methods • the development of products that are culturally acceptable, not offensive to people of different race, gender or religious belief • the development of products that are inclusive • the design and manufacture of products that could assist with social problems, eg poverty, health and wellbeing, migration and housing • the impact of Fairtrade on design and consumer demand • designing products to consider the six Rs of sustainability • the concept of upcycling.	